Simple Harmonic Motion of a Push-Rod

Problem Statement

Practice Ex. 5.10: A push-rod moves with simple harmonic motion driven by an eccentric sheave running at 90 rpm, the full travel of the rod being 50 mm. Calculate its velocity and acceleration when it is 6 mm from the beginning of its travel.

Solution

Given Data

- Speed of the eccentric sheave: $N = 90$ rpm
- Stroke (full travel of the rod): $2A = 50$ mm, so amplitude $A = \frac{50}{2} = 25$ mm $= 0.025$ m
- Position from one end: $x = 6$ mm $= 0.006$ m

Step 1: Angular Velocity

The motion follows simple harmonic motion (SHM), where the angular velocity is:

$$\omega = 2\pi f$$

Since frequency is:

$$f = \frac{N}{60} = \frac{90}{60} = 1.5 \text{ Hz}$$

we get:

$$\omega = 2\pi \times 1.5 = 3\pi \text{ rad/s}$$

Step 2: Displacement from Mid-Travel

Displacement from the mid-position is:

$$x' = A - x = 0.025 - 0.006 = 0.019 \text{ m}$$

Step 3: Phase Angle

Using the SHM equation:

$$\cos\theta = \frac{x'}{A}$$

$$\cos\theta = \frac{0.019}{0.025} = 0.76$$

Solving for θ:

$$\theta = \cos^{-1}(0.76) \approx 40.54^\circ$$

Step 4: Velocity in SHM

Velocity of the push-rod is given by:

$$V_p = \omega A \sin\theta$$

Substituting the values:

$$V_p = (3\pi) \times 0.025 \times \sin 40.54^\circ$$

Approximating:

$$V_p \approx (9.42) \times 0.025 \times 0.651$$

$$V_p \approx 0.153 \text{ m/s} = 153 \text{ mm/s}$$

Step 5: Acceleration in SHM

Acceleration of the push-rod is given by:

$$a_p = \omega^2 A \cos\theta$$

Substituting the values:

$$a_p = (3\pi)^2 \times 0.025 \times 0.76$$

$$a_p = 88.8 \times 0.025 \times 0.76$$

$$a_p \approx 1.69 \text{ m/s}^2 = 1690 \text{ mm/s}^2$$

Final Answers

- Velocity at $x = 6$ mm: **153 mm/s**
- Acceleration at $x = 6$ mm: **1690 mm/s²**

Thus, the push-rod has a velocity of approximately **153 mm/s** and an acceleration of **1690 mm/s²** when it is 6 mm from the beginning of its travel.